Local directory created:

```
Last login: Fri Apr 10 14:59:09 2026 from 157.51.144.137
[ec2-user@ip-172-31-35-127 ~]$ sudo -i
[root@ip-172-31-35-127 ~]# git --version
git version 2.50.1
[root@ip-172-31-35-127 ~]# mkdir vcs-task
[root@ip-172-31-35-127 ~]# cd vcs-task/
[root@ip-172-31-35-127 vcs-task]#
```

Creating Script files:

```
[root@ip-172-31-35-127 vcs-task]# echo "# VCS Task" > README.md
[root@ip-172-31-35-127 vcs-task]# echo "echo 'VCS Task creating local directory with script files'" > script.sh
[root@ip-172-31-35-127 vcs-task]# ls -ltrh
total 8.0K
-rw-r--r--. 1 root root 11 Apr 13 07:49 README.md
-rw-r--r--. 1 root root 59 Apr 13 07:50 script.sh
[root@ip-172-31-35-127 vcs-task]# chmod +x script.sh
[root@ip-172-31-35-127 vcs-task]# ls -ltrh
total 8.0K
-rw-r--r--. 1 root root 11 Apr 13 07:49 README.md
-rwxr-xr-x. 1 root root 59 Apr 13 07:50 script.sh
[root@ip-172-31-35-127 vcs-task]#
```

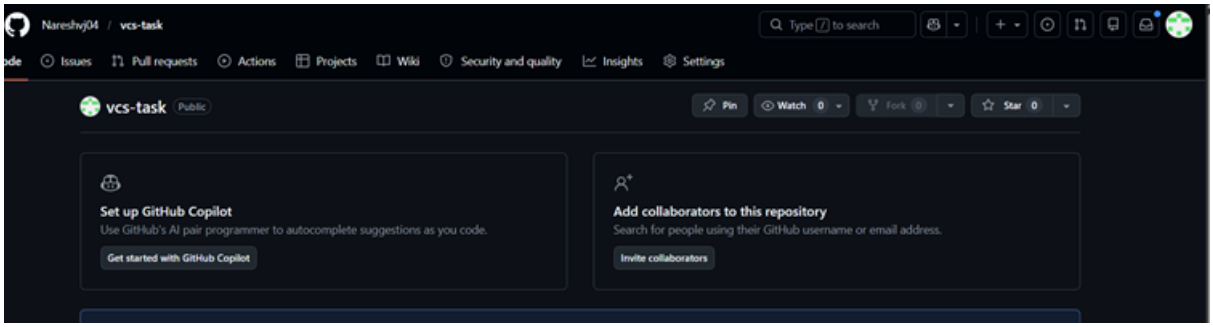
Initialize Git in local directory and added new files:

```
[root@ip-172-31-35-127 vcs-task]# git init -b main
Initialized empty Git repository in /root/vcs-task/.git/
[root@ip-172-31-35-127 vcs-task]# git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        README.md
        script.sh

nothing added to commit but untracked files present (use "git add" to track)
[root@ip-172-31-35-127 vcs-task]# git add .
[root@ip-172-31-35-127 vcs-task]# git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md
        new file:   script.sh
```

Files commit :

```
[root@ip-172-31-35-127 vcs-task]# git commit -m "Initial commit"
[main (root-commit) 63798ed] Initial commit
 Committer: root <root@ip-172-31-35-127.ap-south-1.compute.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 2 files changed, 2 insertions(+)
 create mode 100644 README.md
 create mode 100755 script.sh
[root@ip-172-31-35-127 vcs-task]#
```

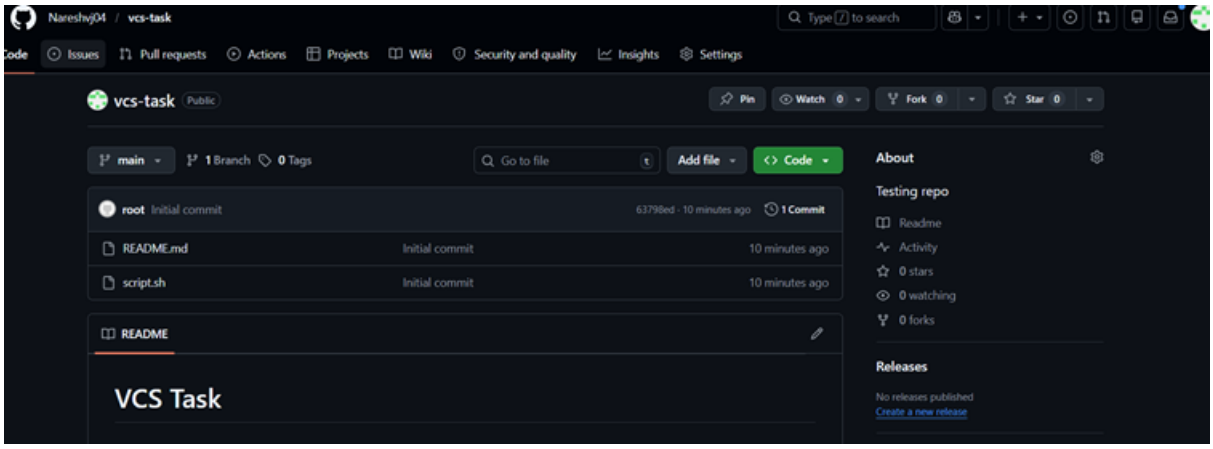
Creating Repository on Github:



Link Local directory to GitHub repo:

```
[root@ip-172-31-35-127 vcs-task]# git remote add origin git@github.com:Nareshvj04/vcs-task.git
[root@ip-172-31-35-127 vcs-task]# git branch -M main
[root@ip-172-31-35-127 vcs-task]# git push -u origin main
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 345 bytes | 345.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Nareshvj04/vcs-task.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
[root@ip-172-31-35-127 vcs-task]#
```

Linked to Github repo:



Creating Feature branch:

```
[root@ip-172-31-35-127 vcs-task]# git checkout -b features
Switched to a new branch 'features'
[root@ip-172-31-35-127 vcs-task]# ls -ltrh
total 8.0K
-rw-r--r--. 1 root root 11 Apr 13 07:49 README.md
-rwxr-xr-x. 1 root root 59 Apr 13 07:50 script.sh
```

Stash:

Using stash to pause current work and switch to another branch for work:

Working file:

```
root@ip-172-31-35-127:~/vcs-
echo 'VCS Task creating local directory with script files'
echo working from main branch
~
~
~
```

Git stash:

```
[root@ip-172-31-35-127 vcs-task]# git stash
Saved working directory and index state WIP on main: 63798ed Initial commit
[root@ip-172-31-35-127 vcs-task]#
```

Switch main branch to Features branch

Modify files from Feature Branch:

```
root@ip-172-31-35-127:~/vcs-
echo 'VCS Task creating local directory with script files'
echo this line added from feature branch
~
~
~
~
~
~
~
~
~
~
```

File commit from Features branch:

```
Saved working directory and index state WIP on main: 63798ed Initial commit
[root@ip-172-31-35-127 vcs-task]# git checkout features
Switched to branch 'features'
[root@ip-172-31-35-127 vcs-task]# vim script.sh
[root@ip-172-31-35-127 vcs-task]# git add .
[root@ip-172-31-35-127 vcs-task]# git commit -m "script modified from feature branch"
[features 7ffd0c1] script modified from feature branch
 Committer: root <root@ip-172-31-35-127.ap-south-1.compute.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 1 insertion(+)
[root@ip-172-31-35-127 vcs-task]#
```

Resume work from main branch:

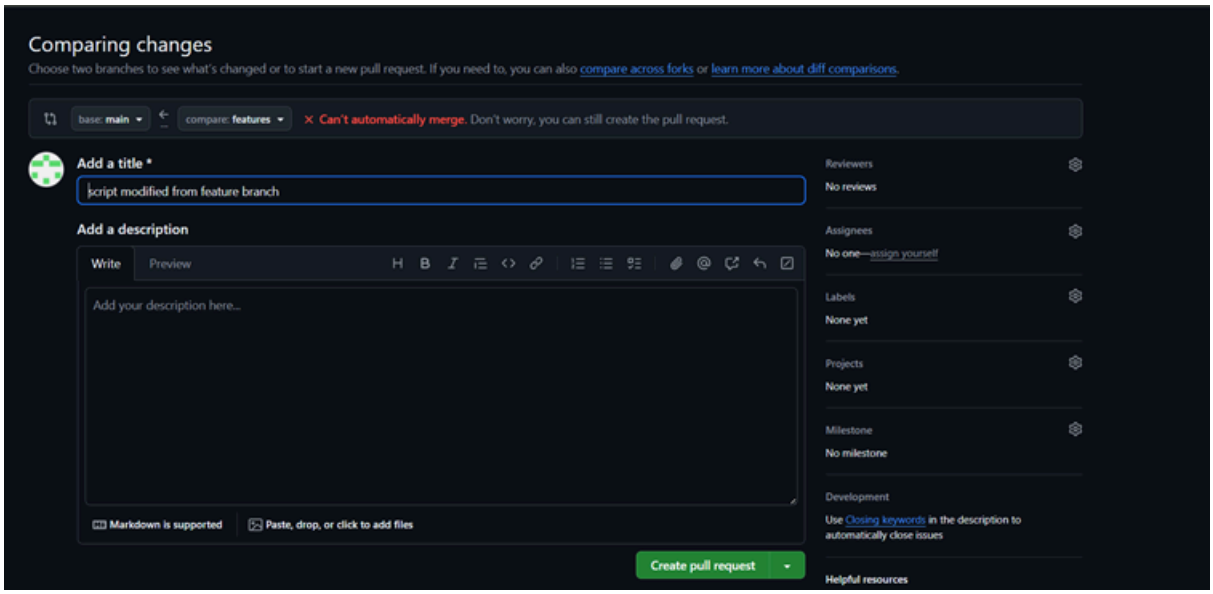
Git stash pop:

```
[root@ip-172-31-35-127 vcs-task]# git stash pop
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   script.sh

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (bb1c5bc9c66c406db2769f713f69ebeaf2581ea2)
[root@ip-172-31-35-127 vcs-task]# vim script.sh
```

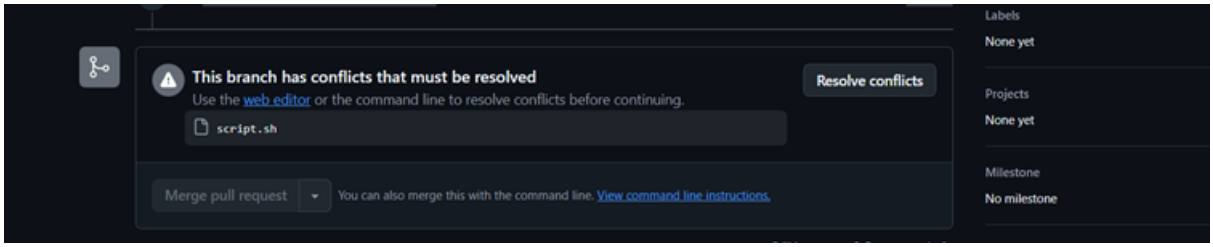
Git Merge:

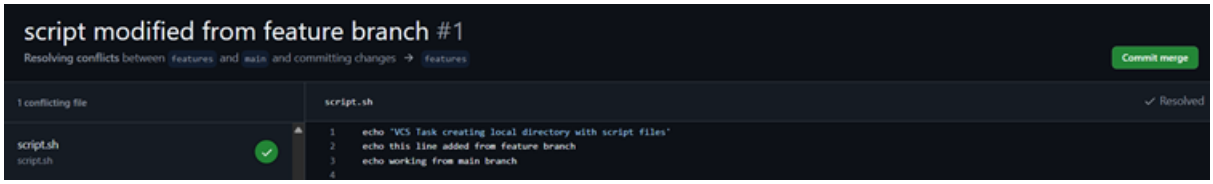
```
[root@ip-172-31-35-127 vcs-task]# git merge
Already up to date.
[root@ip-172-31-35-127 vcs-task]# git merge features
Auto-merging script.sh
CONFLICT (content): Merge conflict in script.sh
Automatic merge failed; fix conflicts and then commit the result.
[root@ip-172-31-35-127 vcs-task]#
```

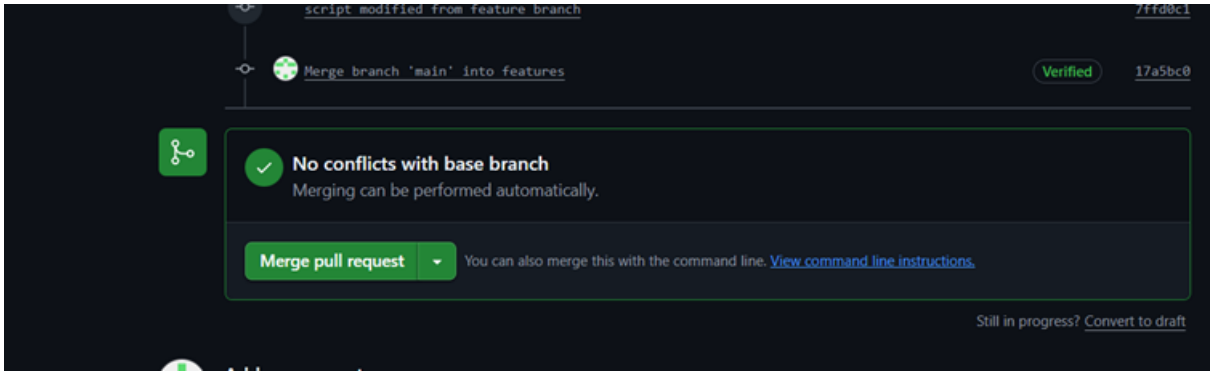
Creating PR:

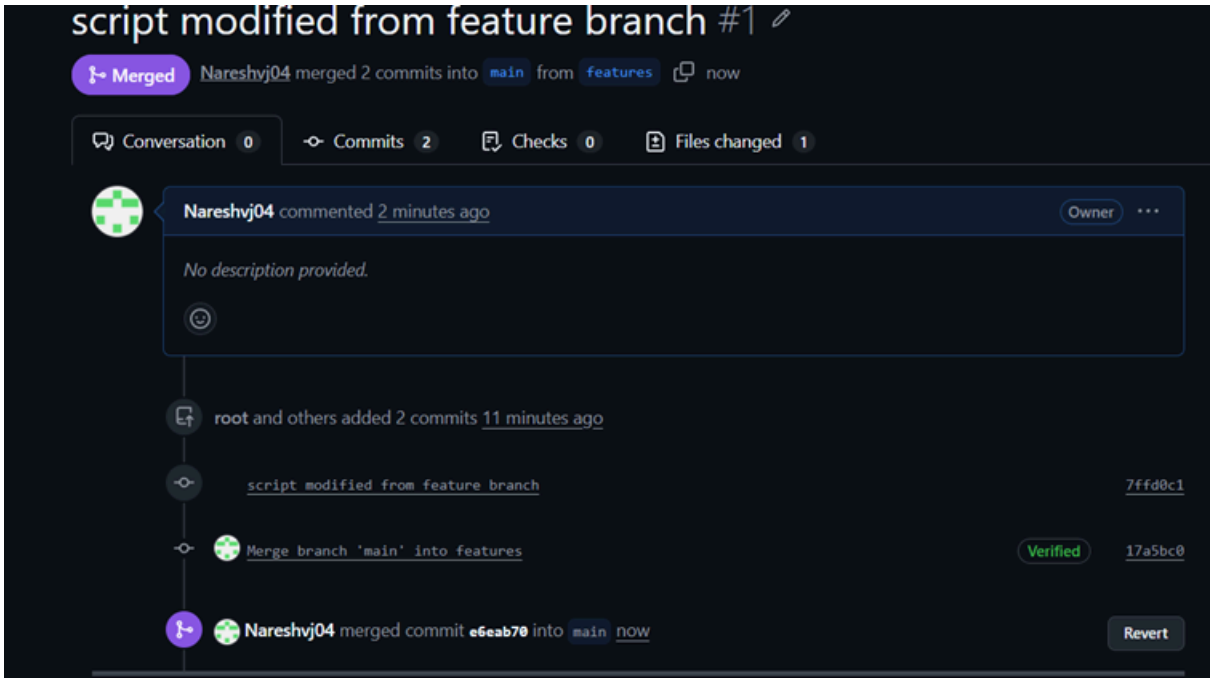


Resolving conflicts:

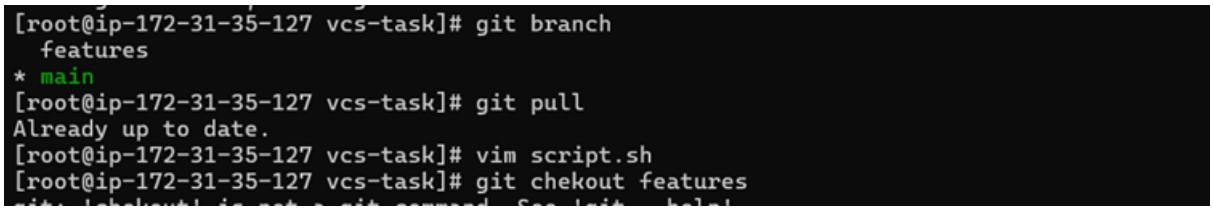








Git pull for updated changes to main:



```
[root@ip-172-31-35-127 vcs-task]# git branch
  features
* main
[root@ip-172-31-35-127 vcs-task]# git pull
Already up to date.
[root@ip-172-31-35-127 vcs-task]# vim script.sh
[root@ip-172-31-35-127 vcs-task]# git chekout features
```

GIT REBASE:

```
[root@ip-172-31-35-127 vcs-task]# git checkout features
Switched to branch 'features'
Your branch is behind 'origin/features' by 2 commits, and can be fast-forwarded.
  (use "git pull" to update your local branch)
[root@ip-172-31-35-127 vcs-task]# git rebase main
Successfully rebased and updated refs/heads/features.
[root@ip-172-31-35-127 vcs-task]# |
```